API_description

LUMI Backend — API Specification

> Base URL: `http://localhost:8000`
> API Prefix: `/api/v1`
> Content-Type: `application/json`
> Docs: `GET /docs` (Swagger UI) | `GET /redoc` (ReDoc)

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Common

Authentication Header

All endpoints except `/auth/*` and `/health` require user identification. During development, auth is bypassed via header:

```
X-User-Id: <uuid>
```

If omitted, defaults to `00000000-0000-0000-0000-000000000001`.

Error Response Format

```
{
  "detail": "Error message"
}
```

Status	Exception	When
401	`UnauthorizedError`	Missing/invalid auth
403	`ForbiddenError`	Insufficient permissions
404	`NotFoundError`	Resource not found
422	`ValidationError`	Invalid input / business rule violation
502	`HealthConnectError`	Upstream sync failure

1. Authentication

1.1 Register

Create a new user account.

```
POST /api/v1/auth/register
```

Request Body

Field	Type	Required	Description
`email`	string (email)	✅	User email address
`chronotype`	string	✅	`"Early Bird"`, `"Night Owl"`, or `"Neutral"`

```
{
  "email": "user@example.com",
  "chronotype": "Night Owl"
}
```

Response `200 OK`

```
{
  "user_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "email": "user@example.com",
  "chronotype": "Night Owl"
}
```

Errors

Status	Condition
422	Email already registered

1.2 Login

Look up an existing user by email.

```
POST /api/v1/auth/login
```

Request Body

Field	Type	Required	Description
`email`	string (email)	✅	Registered email address

```
{
  "email": "user@example.com"
}
```

Response `200 OK`

```
{
  "user_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "email": "user@example.com",
  "chronotype": "Night Owl"
}
```

Errors

Status	Condition
404	User not found

2. Users

2.1 Update SNOP

Manually override the user's Sleep Need for Optimal Performance.

```
PATCH /api/v1/users/me/snop
```

Headers

```
X-User-Id: <uuid>
```

Request Body

Field	Type	Required	Constraints	Description
`snop_hours`	float	✅	4.0 – 12.0	Personalized sleep need in hours

```
{
  "snop_hours": 7.5
}
```

Response `200 OK`

```
{
  "user_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "email": "user@example.com",
  "chronotype": "Night Owl",
  "snop_hours": 7.5
}
```

Side Effects

- Invalidates today's cached energy schedule in Redis.
- Next `GET /energy/schedule` will recompute with the new SNOP value.

Errors

Status	Condition
404	User not found
422	`snop_hours` outside 4.0–12.0 range

3. Survey

3.1 Submit Onboarding Survey

Submit survey answers for SNOP seed calibration (FR-01).

```
POST /api/v1/survey/submit
```

Headers

```
X-User-Id: <uuid>
```

Request Body

Field	Type	Required	Description
`answers`	array	✅	List of question/answer pairs

Each answer object:

Field	Type	Required	Description
`question_id`	string	✅	Question identifier (e.g. `"age"`, `"gender"`, `"sleep_schedule"`)
`answer_key`	string	✅	Answer value (e.g. `"30"`, `"M"`, `"7-8h"`)

```
{
  "answers": [
    { "question_id": "age", "answer_key": "30" },
    { "question_id": "gender", "answer_key": "M" },
    { "question_id": "sleep_schedule", "answer_key": "7-8h"
},
    { "question_id": "goal", "answer_key": "performance" },
    { "question_id": "health_issues", "answer_key": "none"
}
  ]
}
```

Response `200 OK`

```
[
  { "user_id": "a1b2c3d4-...", "question_id": "age", "answer_key": "30" },
  { "user_id": "a1b2c3d4-...", "question_id": "gender", "answer_key": "M" },
  { "user_id": "a1b2c3d4-...", "question_id": "sleep_schedule", "answer_key": "7-8h" },
  { "user_id": "a1b2c3d4-...", "question_id": "goal", "answer_key": "performance" },
  { "user_id": "a1b2c3d4-...", "question_id": "health_issues", "answer_key": "none" }
]
```

Side Effects

- Stores all answers in `user_survey_responses` table.
- Triggers SNOP seed calibration from actigraphy reference data (age + gender + self-reported schedule).

---

4. Sleep

4.1 Health Connect Sync

Ingest bulk sleep and biometric data from Android Health Connect (FR-02).

```
POST /api/v1/sleep/sync
```

Headers

```
X-User-Id: <uuid>
```

Request Body

Field	Type	Required	Description
`sleep_sessions`	array	✅	List of sleep session records
`heart_rate_records`	array	❌	List of heart rate data points
`steps_records`	array	❌	List of step count data points

Each sleep session:

Field	Type	Required	Description
`start_time`	datetime	✅	Sleep start (ISO 8601)
`end_time`	datetime	✅	Wake up time (ISO 8601)
`duration_mins`	integer	✅	Actual sleep duration in minutes
`session_type`	string	❌	`"nightly"` or `"nap"` — auto-classified if omitted

Each heart rate record:

Field	Type	Description
`time`	string (ISO 8601)	Measurement timestamp
`value`	number	Heart rate in BPM

Each steps record:

Field	Type	Description
`time`	string (ISO 8601)	Measurement timestamp
`value`	number	Step count

```
{
  "sleep_sessions": [
    {
      "start_time": "2026-05-29T23:00:00+07:00",
      "end_time": "2026-05-30T06:30:00+07:00",
      "duration_mins": 450
    }
  ],
  "heart_rate_records": [
    { "time": "2026-05-30T02:15:00+07:00", "value": 58.5 },
    { "time": "2026-05-30T04:30:00+07:00", "value": 62.0 }
  ],
  "steps_records": [
    { "time": "2026-05-30T14:00:00+07:00", "value": 8500 }
  ]
}
```

Response `200 OK`

```
{
  "sessions_dispatched": 1,
  "biometrics_dispatched": 3
}
```

Behavior

- Returns immediately (non-blocking).
- Sleep sessions are sanitized: sessions < 5 min are rejected, type is auto-classified if not provided (10 AM – 6 PM = nap, otherwise nightly).
- Data is dispatched to Celery storage workers for async DB insert.
- After storage completes, energy recalculation is triggered automatically.

4.2 Manual Sleep Input

Manual sleep entry for users without Health Connect (FR-02 graceful degradation).

```
POST /api/v1/sleep/manual
```

Headers

```
X-User-Id: <uuid>
```

Request Body

Field	Type	Required	Constraints	Description
`start_time`	datetime	✅	—	Sleep start (ISO 8601)
`end_time`	datetime	✅	—	Wake up time (ISO 8601)
`duration_mins`	integer	✅	≥ 5	Actual sleep duration in minutes
`session_type`	string	❌	`"nightly"` or `"nap"`	Auto-classified if omitted

```
{
  "start_time": "2026-05-30T13:00:00+07:00",
  "end_time": "2026-05-30T13:25:00+07:00",
  "duration_mins": 25
}
```

Response `200 OK`

```
{
  "sessions_dispatched": 1,
  "session_type": "nap"
}
```

Errors

Status	Condition
200 with error	`duration_mins` < 5 → `{"error": "Session too short (minimum 5 minutes)", "sessions_dispatched": 0}`

4.3 List Sleep Sessions

Retrieve recent sleep sessions.

```
GET /api/v1/sleep/sessions
```

Headers

```
X-User-Id: <uuid>
```

Query Parameters

Param	Type	Default	Description
`limit`	integer	14	Max number of sessions to return

```
GET /api/v1/sleep/sessions?limit=7
```

Response `200 OK`

```
[
  {
    "session_id": "b2c3d4e5-...",
    "user_id": "a1b2c3d4-...",
    "start_time": "2026-05-29T23:00:00+07:00",
    "end_time": "2026-05-30T06:30:00+07:00",
    "duration_mins": 450,
    "session_type": "nightly"
  },
  {
    "session_id": "c3d4e5f6-...",
    "user_id": "a1b2c3d4-...",
    "start_time": "2026-05-30T13:00:00+07:00",
    "end_time": "2026-05-30T13:25:00+07:00",
    "duration_mins": 25,
    "session_type": "nap"
  }
]
```

Ordered by `start_time` descending (most recent first).

4.4 Get Sleep Debt

Compute cumulative sleep debt over a 14-day sliding window (FR-03).

```
GET /api/v1/sleep/debt
```

Headers

```
X-User-Id: <uuid>
```

Response `200 OK`

```
{
  "user_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "sleep_debt_mins": 270.0,
  "sleep_debt_display": "4h 30m",
```

```
  "snop_hours": 8.0,
  "window_days": 14
}
```

Field	Type	Description
`sleep_debt_mins`	float	Total debt in minutes (0 if user is not in debt)
`sleep_debt_display`	string	Human-readable format (e.g. `"4h 30m"`, `"0h 0m"`)
`snop_hours`	float	Current personalized SNOP
`window_days`	int	Always 14 (sliding window size)

Calculation

```
Sleep Debt = Σ (SNOP - ActualSleep_i) for i = 1..14 days
```

- `ActualSleep_i` includes both nightly sleep and naps per calendar day.
- Debt is floored at 0 (no negative debt).
- Warning threshold: 300 minutes (5 hours).

---

5. Energy

5.1 Get Energy Schedule

Compute or retrieve cached 24-hour energy schedule (FR-04, FR-05).

```
GET /api/v1/energy/schedule
```

Headers

```
X-User-Id: <uuid>
```

Query Parameters

Param	Type	Default	Description
`date`	datetime	today	Date to compute schedule for (ISO 8601)

```
GET /api/v1/energy/schedule?date=2026-05-30T00:00:00Z
```

Response `200 OK`


```
{
  "user_id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "date": "2026-05-30T00:00:00Z",
  "energy_points": [
    { "time": "2026-05-30T07:00:00+07:00", "value": 35.2 },
    { "time": "2026-05-30T07:05:00+07:00", "value": 38.1 },
    "...288 points total (5-min resolution over 24h)..."
  ],
  "zones": [
    {
      "zone_type": "wake",
      "start_time": "2026-05-30T07:00:00+07:00",
      "end_time": "2026-05-30T08:30:00+07:00"
    },
    {
      "zone_type": "morning_peak",
      "start_time": "2026-05-30T10:00:00+07:00",
      "end_time": "2026-05-30T11:00:00+07:00"
    },
    {
      "zone_type": "afternoon_dip",
      "start_time": "2026-05-30T14:00:00+07:00",
      "end_time": "2026-05-30T16:00:00+07:00"
    },
    {
      "zone_type": "evening_peak",
      "start_time": "2026-05-30T18:00:00+07:00",
      "end_time": "2026-05-30T19:00:00+07:00"
    },
    {
      "zone_type": "melatonin_window",
      "start_time": "2026-05-30T22:00:00+07:00",
      "end_time": "2026-05-31T00:00:00+07:00"
```

```
    }
  ],
  "nudges": [
    {
      "time": "2026-05-30T07:00:00+07:00",
      "message": "Get 10-15 min of bright natural light to
clear sleep inertia and reset your circadian clock",
      "nudge_type": "light_exposure"
    },
    {
      "time": "2026-05-30T07:10:00+07:00",
      "message": "Drink a full glass of water — your body i
s dehydrated after sleep",
      "nudge_type": "hydration"
    },
    "...16 nudges total..."
  ],
  "energy_potential": 72.5,
  "is_cached": false
}
```


Response Fields

Field	Type	Description
`energy_points`	array	288 data points, 5-min resolution, values 0–100
`zones`	array	5 physiological zones with start/end times
`nudges`	array	16 science-based behavioral nudges
`energy_potential`	float	0–100 score. Higher = more peak energy capacity
`is_cached`	bool	`true` if served from Redis cache, `false` if freshly computed

Energy Potential Score

```
score = (1 - h_at_wake) × 100
where h_at_wake = min(sleep_debt / SNOP, 1.0)
```

Score	Meaning
90–100	Excellent — full energy capacity
70–89	Good — mild debt impact
50–69	Moderate — noticeable energy reduction
30–49	Low — significant fatigue expected
0–29	Critical — severe exhaustion

Zone Types

Zone	Description	Duration
`wake`	Sleep inertia period	0–90 min after waking
`morning_peak`	Highest cognitive performance	~60 min window
`afternoon_dip`	Natural energy low	~120 min window
`evening_peak`	Second energy rise	~60 min window
`melatonin_window`	DLMO — time to prepare for sleep	~120 min, dynamic per user

Nudge Types

Type	Example Message
`light_exposure`	Get 10-15 min of bright natural light...
`hydration`	Drink a full glass of water...
`exercise`	Best time for exercise — physical activity now boosts alertness...
`deep_work`	Peak alertness window — schedule your most important deep work now
`caffeine`	Ideal window for caffeine — pair with your natural cortisol peak...
`caffeine_cutoff`	Caffeine cutoff — stop all caffeine now to protect tonight's sleep
`meal_timing`	Finish any heavy meals now — digestion too close to bed disrupts sleep
`nap`	Energy dip detected — a 20-min power nap now can reduce sleep debt...
`task_suggestion`	Low energy window — switch to routine tasks, emails, or light meetings
`blue_light`	Start reducing blue light — enable night mode, dim screens...
`wind_down`	Melatonin production starting — begin your wind down: light reading...

Caching

- Results are cached in Redis with 24-hour TTL, keyed by `energy:{user_id}:{YYYY-MM-DD}`.
- If cached, returns immediately with `is_cached: true`.
- Cache is invalidated when SNOP is manually updated via `PATCH /users/me/snop`.

Computation Pipeline (when cache miss)

1. Fetch last 14 days of sleep sessions
2. Aggregate daily sleep totals (nightly + naps)
3. Compute cumulative sleep debt
4. Estimate circadian phase (phi) from recent bedtimes/wake times
5. Compute exertion-adjusted SNOP from heart rate data
6. Compute Process S at wake time
7. Run Borbély two-process model synthesis (Process S + Process C)
8. Classify 5 energy zones
9. Generate 16 science-based nudges
10. Compute Energy Potential Score
11. Cache result in Redis

---

6. Health Check

```
GET /health
```

Response `200 OK`

```
{
  "status": "ok"
}
```

No authentication required.

Summary Table

Method	Endpoint	Auth	Description
`GET`	`/health`	❌	Health check
`POST`	`/api/v1/auth/register`	❌	Register new user
`POST`	`/api/v1/auth/login`	❌	Login by email
`PATCH`	`/api/v1/users/me/snop`	✅	Manual SNOP override
`POST`	`/api/v1/survey/submit`	✅	Submit onboarding survey
`POST`	`/api/v1/sleep/sync`	✅	Health Connect bulk sync
`POST`	`/api/v1/sleep/manual`	✅	Manual sleep entry
`GET`	`/api/v1/sleep/sessions`	✅	List recent sleep sessions
`GET`	`/api/v1/sleep/debt`	✅	Get current sleep debt
`GET`	`/api/v1/energy/schedule`	✅	Get 24h energy schedule

Total: 10 endpoints